Tal Simhayev

Looking for a Security Research position. I'm a self-learner who was trained by military cyber experts. I participate in CTFs and have created multiple research projects on Windows and Linux platforms.
See all my projects on my website: https://talsim.github.io

Personal Details

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Experience

- February 2024 - Present – Software Engineer, Israel Defense Forces
  - Collaborated in an R&D team to develop automation and ease software workflows using Playwright.
  - Integrated automated processes to the team's CI/CD pipelines, with a focus on efficiency in deployments.

- Participated in CTF events with my team.
  - Published writeups for Flare-On 2024.
  - Published on **CTFtime:** pingCTF 2023 - dangle-me pwn challenge writeup
  - **See my blog post in Medium**: ROP Emporium write4 - by Tal Simhayev
  - See more writeups in my CTFs Github Repo.

- Extensive hands-on experience working with static and dynamic binary analysis tools including:
  - IDA Pro, Ghidra and Radare2 disassemblers.
  - Dynamic and Static debugging (e.g. GDB, WinDbg, OllyDbg).
  - Knowledge in x86/x64 Assembly language.
  - Vulnerability research - e.g. exploiting DirtyCow vulnerability in the Linux kernel to gain root privileges github.com/talsim/root-dirtyc0w.

- Experience with C/C++ languages including:
  - Windows Internals - Developing a **Remote Access Trojan (RAT)** (github.com/talsim/CloakRAT, work in progress), focusing on PE injection, persistence, anti-debugging, and antivirus evasion.
  - Linux Internals (implementing Linux shell - github.com/talsim/Linux-Shell).
  - Writing a KeyLogger using **WinAPI** - github.com/talsim/KeyLogger.

- Extensive networks knowledge:
  - TCP/IP model, working with Sockets, Server-Client architecture, HTTP server writing, good understanding of network protocols - UDP, TCP, DNS, DHCP, etc. (github.com/talsim/Networks)
  - Network traffic analysis with Wireshark software, and usage of Scapy package in Python.

Education

2016 - 2022
- Student at ORT-Rogozin High School, Kiryat Gat. Major in Computer Science - 10 points (score 100) & Physics.

Community Involvements

2019 - Young Initiators in Education project by the Center for Educational Technology:
- Participated actively and brought up ideas on how to improve mathematics studies in Israel middle schools with an emphasis on personalized learning.